

Repeated Games

Two firms set prices today and meet again tomorrow. A choice today can change how the other firm behaves tomorrow. Fix a normal-form game, called the *stage game*, and play it in periods $t = 0, 1, 2, \dots$. At the end of each period everyone observes the actions played.

In a one-shot game, a strategy tells a player what to do in that game. Here it must also say how to respond to earlier play: for example, cooperate today if everyone has cooperated so far, and defect otherwise.

Definition 1. A *history* h_t records everyone's actions in periods 0 through $t - 1$; h_0 is empty. A *strategy* in the repeated game assigns a stage-game action (or distribution over actions) to every history.

Definition 2. Players have a common *discount factor* $\delta \in (0, 1)$. Player i evaluates a stream of stage payoffs by

$$\sum_{t \geq 0} \delta^t u_i^t,$$

with expectation over any randomization. Stage payoffs are bounded.

A payoff of 1 tomorrow is worth δ today; the same payoff two periods from now is worth δ^2 . Larger δ means that future payoffs matter more. We use unnormalized discounted payoffs: we do not multiply the sum by $1 - \delta$.

What is the value of receiving u every period? Of receiving d now and w every period starting tomorrow?

Discounted values.

$$\begin{aligned} \sum_{t \geq 0} \delta^t u &= u(1 + \delta + \delta^2 + \dots) = \frac{u}{1 - \delta}, \\ d + \sum_{t \geq 1} \delta^t w &= d + \delta w(1 + \delta + \delta^2 + \dots) = d + \frac{\delta w}{1 - \delta}. \end{aligned}$$

Subgame Perfection

Nash equilibrium rules out a beneficial unilateral change to a player's whole strategy. But a threat can deter deviations even if the player making it would not want to carry it out. If the threatened history is never reached, changing the action there does not improve that player's payoff from the start of the game.

A threatened punishment must itself be optimal when the time comes to carry it out.

Definition 3. A *subgame* is the repeated game remaining after a history h_t . A strategy profile is a *subgame-perfect equilibrium* (SPE) if it induces a Nash equilibrium after every history, including histories that never occur when everyone follows the profile.

Complete the comparison:

- Nash equilibrium checks deviations starting at the initial history.
- Subgame perfection checks deviations starting after every history.

Theorem 1 (One-shot deviation principle). *In this discounted repeated game, a strategy profile is subgame perfect if and only if, after every history, no player can gain by changing their action in one period and following the prescribed strategy thereafter.*

We take the principle as given. “Thereafter” means following the strategy at the history produced by the deviation, including any punishment it triggers. We can therefore compare today’s payoff plus the continuation payoff, instead of checking every possible change to a whole strategy.

Grim Trigger

Payoffs are listed as (row player, column player).

	Column cooperates	Column defects
Row cooperates	(3, 3)	(0, 4)
Row defects	(4, 0)	(1, 1)

Find the stage-game equilibrium and each player’s payoff there.

Stage equilibrium. Defect strictly dominates Cooperate, so (Defect, Defect) is the unique stage equilibrium, paying each player 1.

Both players would receive more from (Cooperate, Cooperate). To sustain that outcome, today’s action must affect tomorrow’s play. Consider the following strategy.

Grim trigger prescribes a target action profile as long as everyone has always played it. After any deviation, it prescribes a fixed Nash equilibrium of the stage game forever.

At any history, compare payoffs from that period onward, treating it as today. Earlier payoffs are fixed, and the common discount factor multiplying all remaining payoffs does not change which continuation is better.

Exercise. For which discount factors does grim trigger sustain (Cooperate, Cooperate) as an SPE? Check both histories with a past deviation and histories without one.

Proof. After a past deviation. Future play is Defect regardless of the current action. Defect is a best response today, so there is no beneficial one-shot deviation.

Without a past deviation. Fix the column player's grim-trigger strategy. The row player receives 3 each period by conforming. Defecting pays the row player 4 today and triggers payoff 1 forever after. The same comparison holds for the column player. The comparison is

$$\underbrace{\frac{3}{1-\delta}}_{\text{cooperate forever}} \geq \underbrace{4}_{\text{defect today}} + \underbrace{\frac{\delta}{1-\delta}}_{\text{punishment thereafter}}.$$

Discount factor. Since $1 - \delta > 0$,

$$3 \geq 4(1 - \delta) + \delta \iff 3\delta \geq 1 \iff \delta \geq \frac{1}{3}.$$

For these discount factors neither player has a beneficial one-shot deviation after either kind of history, so the profile is SPE. For $\delta < 1/3$, Defect is a beneficial deviation at the initial history, so it is not SPE. $\square$

What happens to this incentive comparison as $\delta \rightarrow 1$ and as $\delta \rightarrow 0$?

Patience. For δ near 1, losing the future cooperation payoff outweighs the immediate gain. For δ near 0, the immediate gain determines the choice. At $\delta = 1/3$, conforming and deviating give the same value.

The argument used the following steps.

Step 1: Find the punishment equilibrium. Check that repeating it makes the punishment credible.

Step 2: Name the target and its per-period payoff. Compute each player's payoff from conforming.

Step 3: Compute the best one-shot deviation. Hold the other players' current actions fixed.

Step 4: Write down conform $\geq$ deviate. Include the continuation payoff on each side.

Step 5: Solve for δ . Check the condition for every player.

General Threshold (Optional Extension)

Exercise. Now fix a player and let the target payoff be R , the best one-period deviation payoff be D , and the punishment equilibrium payoff be P , where $D > R > P$. Derive the discount-factor threshold.

General threshold.

$$\begin{aligned} \frac{R}{1-\delta} &\geq D + \frac{\delta P}{1-\delta} \\ R &\geq D(1-\delta) + \delta P \\ R &\geq D - \delta(D-P) \\ \delta &\geq \frac{D-R}{D-P}. \end{aligned}$$

The condition must hold for every player, using that player's R, D, P . Holding D and R fixed, what happens when P decreases? Why can't we simply choose any small P we want?

	Column cooperates	Column defects
Row cooperates	(3, 3)	(0, 4)
Row defects	(4, 0)	(1, 1)

Lowering P increases the denominator and lowers the threshold. But the punishment must be an equilibrium continuation. In our matrix, threatening (Cooperate, Defect) forever gives the row player $P = 0$, but that player can increase today's payoff to 1 by Defecting. This punishment is not credible.

Finite Horizons

Suppose instead there are exactly T periods, numbered $0, \dots, T - 1$, and everybody knows this.

Claim 1. If the stage game has a unique Nash equilibrium (including mixed strategies), every SPE of its T -period repetition prescribes that equilibrium after every history.

Why must the same equilibrium be played after every history?

Proof. In period $T - 1$ there is no future, so after any history play must be the unique stage equilibrium. Its payoff is therefore independent of what happened earlier. In period $T - 2$, a current deviation cannot change the continuation payoff, so play must again be the stage equilibrium. Repeat this argument backward to period 0. $\square$

What does the claim imply for ten repetitions of our matrix? Which assumption would fail in a stage game with two Nash equilibria?

Finite repetition. Both players Defect after every history in all ten periods. With multiple stage equilibria, the final-period equilibrium could depend on earlier actions, so its continuation payoff need not be fixed. The backward-induction argument above would no longer apply.

Two Periods, Multiple Equilibria

Now consider a different stage game. The row and column players each choose A, B , or C . Payoffs are listed as (row player, column player).

	Column A	Column B	Column C
Row A	(2, 2)	(0, 3)	(0, 0)
Row B	(3, 0)	(1, 1)	(0, 0)
Row C	(0, 0)	(0, 0)	(3, 3)

Find all pure Nash equilibria of the stage game. Is (A, A) a Nash equilibrium?

Stage equilibria. The pure Nash equilibria are (B, B) and (C, C) , paying each player 1 and 3, respectively. Against A or B , the unique best response is B ; against C , it is C . At (A, A) , either the row or the column player can increase their payoff from 2 to 3 by choosing B .

Exercise. Play this game for exactly two periods, $t = 0, 1$, with payoff $u_i^0 + \delta u_i^1$ and $\delta = 3/4$. Construct an SPE in which both players choose A in period 0. Specify period-1 play after every possible period-0 outcome, and check both players' incentives in both periods.

Strategy. Both players choose A in period 0. In period 1, play (C, C) if the period-0 outcome was (A, A) , and play (B, B) after every other outcome.

Proof. Last period. After every history, the prescribed play is a stage-game Nash equilibrium. Neither player can benefit by changing their last-period action.

First period. Fix the column player's strategy. If the row player chooses A , the row player's payoff is

$$2 + \delta \cdot 3 = \frac{17}{4}.$$

Choosing B gives 3 now and triggers (B, B) tomorrow, for

$$3 + \delta \cdot 1 = \frac{15}{4}.$$

Choosing C gives 0 now and triggers the same punishment, for $\delta = 3/4$. Neither deviation is beneficial. The column player faces the same comparisons. Changing the last-period action as well cannot improve these deviation payoffs, since the prescribed last-period action is already a best response after every history. Thus the strategies are a Nash equilibrium from the start and after every period-0 history, so they form an SPE. $\square$

Which step of the finite-horizon proof no longer works?

History-dependent continuation. Last-period play must still be a stage equilibrium, but its payoff can now depend on first-period actions. The loss of $3 - 1 = 2$ tomorrow deters a gain of $3 - 2 = 1$ today when $2\delta \geq 1$. In fact, this construction works exactly when $\delta \geq 1/2$. A known last period does not by itself force stage-equilibrium play in earlier periods; uniqueness was essential to our claim.

Acknowledgements

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References

Tim Roughgarden. *Twenty Lectures on Algorithmic Game Theory*. Cambridge University Press, 2016.